

WEEK – 4

a) Create an application to implement Decision-making using Dart.

b) Create an application to implement loops using Dart.

4. a) Create an application to implement Decision-making using Dart.

AIM: To implement decision-making statements using Dart.

DECISION-MAKING STATEMENTS:

- **Decision-making statements** are used to **control the flow of program execution** based on a condition.
- These statements evaluate a condition (true or false) and execute a specific block of code accordingly.
- The Dart language supports the following types of decision-making statements:
 1. if Statement
 2. if-else Statement
 3. else-if Ladder
 4. Nested if Statement
 5. switch Statement
 6. Ternary Operator (?:)

The following is the Dart Program to Print Grades using if statement and to print a day using Switch statement:

PROGRAM: *[week4a.dart]*

```
import 'dart:io'; //import input/output dart package
void main()
{
  print("=== Decision Making Demo ===");
  //Read marks
  print("Enter Marks:");
  int?marks=int.parse(stdin.readLineSync());

  print("The Entered Student Marks are: $marks");
  // if-else ladder for grading logic
  if (marks >= 90)
    print("Grade: A+");
  else if (marks >= 75)
    print("Grade: A");
  else if (marks >= 60)
    print("Grade: B");
```

```

else if (marks >= 40)
    print("Grade: C");
else print("Grade: F (Fail)");

// switch-case for day of the week
// Read Day
print("\n Enter 1-7 to Print the Day:");
int?day = int.parse(stdin.readLineSync());
print("\n Day Number: $day");
switch (day)
{
    case 1:
        print("Monday");
        break;
    case 2:
        print("Tuesday");
        break;
    case 3:
        print("Wednesday");
        break;
    case 4:
        print("Thursday");
        break;
    case 5:
        print("Friday");
        break;
    case 6:
        print("Saturday");
        break;
    case 7:
        print("Sunday");
        break;
    default:
        print("Invalid day number");
}

```

OUTPUT:

=== Decision Making and Loops Demo ===

Enter Marks: 83

The Entered Student Marks are: 83

Grade: A

Enter 1-7 to Print the Day: 4

Day Number: 4

Thursday

4. b) Create an application to implement loops using Dart.

AIM: To implement decision-making and looping statements using Dart.

LOOPING STATEMENTS:

- **Loops** are used to execute a block of code repeatedly until a specified condition becomes true.
- Dart language supports the following types of loop statements:
 1. for
 2. while
 3. do-while
 4. for-in
 5. forEach()

Syntaxes:

1. for loop:

```
for(initialization; condition; increment/decrement)
{
    // statements
}
```

2. while loop:

```
while(condition)
{
    // statements
}
```

3. do-while loop:

```
do
{
    // statements
} while(condition);
```

4. for .. in loop:

It is a special type of loop in Dart used to iterate (traverse) through each element of a collection, such as a List, Set, Map (keys or values), or any Iterable object.

Why Do We Use for-in Loop?

The for-in loop is useful when:

- we want to access every element in a collection.
- we do not need the index (position) of the elements.
- we want shorter and more readable code.

Syntax:

```
for(dataType variable in collection)
{
    // Statements
}
```

Example: Printing numbers

```
void main() {
    List<int> numbers = [10,20,30,40,50];
    for(int number in numbers)
    {
        print(number);
    }
}
```

Output: 10

20

30

40

50

Advantages of for-in Loop:

1. Simple and easy to write.
2. No index variable is required.
3. Automatically traverses all elements.
4. Improves code readability.
5. Suitable for Lists, Sets, Maps, and other Iterables.

5. forEach() Loop: It is a method used to process every element in a collection such as a list.

Syntax:

```
collection.forEach((element)
{
    // statements
}
);
```

Example:

```
void main()
{
    List<int> numbers = [10, 20, 30];

    numbers.forEach((number)
    {
        print(number);
    })
};
```

The following is the Dart Program to Print to print a sequence of numbers:

PROGRAM: *[week4b.dart]*

```
void main()
{
    // ----- LOOPS -----
    print("\n FOR loop: Printing Numbers from 1 to 5");
    for (int i = 1; i <= 5; i++)
    {
        print("Number: $i");
    }

    print("\n WHILE loop: Counting down from 5");

    int count = 5;
    while (count > 0)
    {
        print(count);
        count--;
    }

    print("\n DO-WHILE loop: At least one execution");
    int num = 1;

    do
    {
        print("Iteration: $num");
        num++;
    } while (num <= 3);

    print("\n FOR..IN loop: Printing all the elements in the list");

    List<int> numbers = [10,20,30];
    for(int number in numbers)
    {
        print(number);
    }
}
```

```

    print("\n forEach() loop: Printing all the elements in the list");

    List<int> numbers = [10, 20, 30];

    numbers.forEach((number)
    {

        print(number);

    })

} //end of main

```

OUTPUT:

FOR loop: Printing Numbers from 1 to 5

Number: 1

Number: 2

Number: 3

Number: 4

Number: 5

WHILE loop: Counting down from 5

5

4

3

2

1

DO-WHILE loop: At least one execution

Iteration: 1

Iteration: 2

Iteration: 3

FOR..IN loop: Printing all the elements in the list

10

20

30

forEach() loop: Printing all the elements in the list

10

20

30